## Abstract

Public land-cover products are often used where authoritative local time series are unavailable, but product choice and temporal discontinuities can alter apparent urban change and model rankings. We present an auditable protocol for constrained spatial allocation using Geospatial Kernel, the execution layer of a Geospatial World Model. Annual 10-m Google Dynamic World (2017–2024) and ArcGIS-served Impact Observatory/Microsoft/Esri (2017–2025) products for Abu Dhabi city were harmonized to a 100-m, six-class contract. Leakage-controlled expanding-window tests compared a FLUS-style ANN–CA, Geospatial Kernel, GeoFM-LDN, persistence and random minimum-change allocation, using three seeds and 1,000 spatial-block resamples. Target-year labels supplied oracle class totals and evaluation only. In 2024, ArcGIS mapped 335.57 km² built versus 155.98 km² in Dynamic World (built IoU 0.449). Its built stock rose 30.6% from 2021 to 2022 while wetland and woody classes subsequently collapsed, indicating an unreconciled product-series break rather than verified city change. Kernel had the highest mean FoM in all four Dynamic World targets and two matched ArcGIS targets, without universal paired-interval separation. Excluding the discontinuity target, ArcGIS-minus-Dynamic World FoM was negative for Kernel (-0.244 to -0.014) and GeoFM-LDN (-0.215 to -0.010), whereas FLUS ranged from -0.100 to +0.023. Exact minimum-change allocators faced product-specific FoM upper bounds of 0.086–0.813 (0.086–0.665 excluding 2022). Product-specific 2031 frontiers retained FLUS-style and Kernel candidates but changed objective trade-offs. The results support an inspectable proposal–projection–writeback boundary while showing comparative skill is label-product dependent. Neither product is authoritative local land-use truth, and 2026–2031 maps are 100-m conditional stress tests.